PLP - Programmable Load Project

UA-DETI-EA

Datasheet version 1.0
Hardware Rev A. Firmware v1.0.
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Input voltage	0–30 V
Maximum current	5 A
Maximum power	150 W
Supply voltage	12 V
Interface	SPI
Operating modes	CC, CR, CP

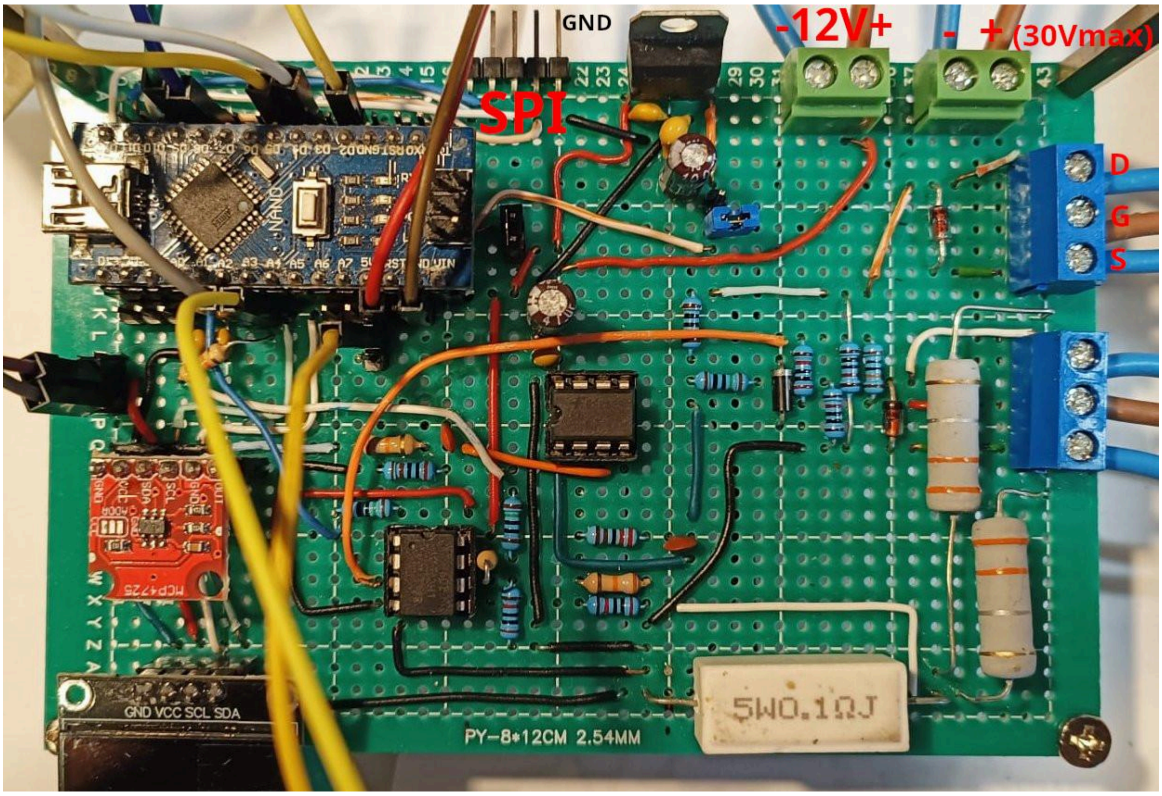

Figure 1 – Prototype main board.

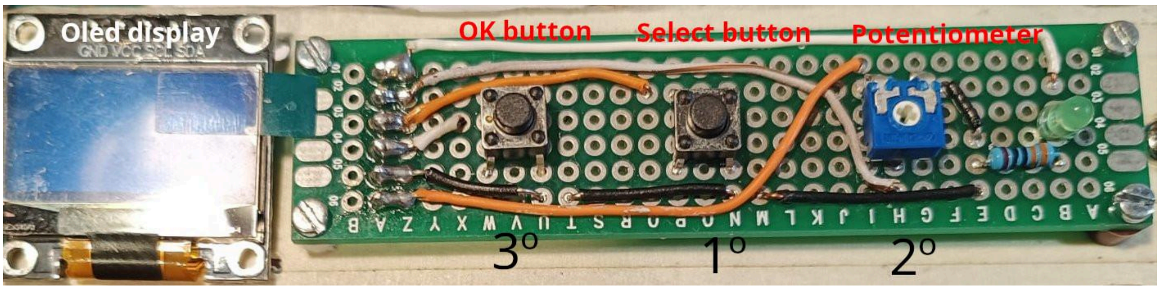

Figure 2 – Prototype human-device interface (HDI) with display and control board.

Overview

The EA Programmable Load is a microcontroller-based DC electronic load capable of operating in Constant Current (CC), Constant Resistance (CR) and Constant Power (CP) modes. The instrument is intended for educational use, power supply characterization and photovoltaic panel testing, providing adjustable load control and real-time measurement of voltage and current.

Block diagram

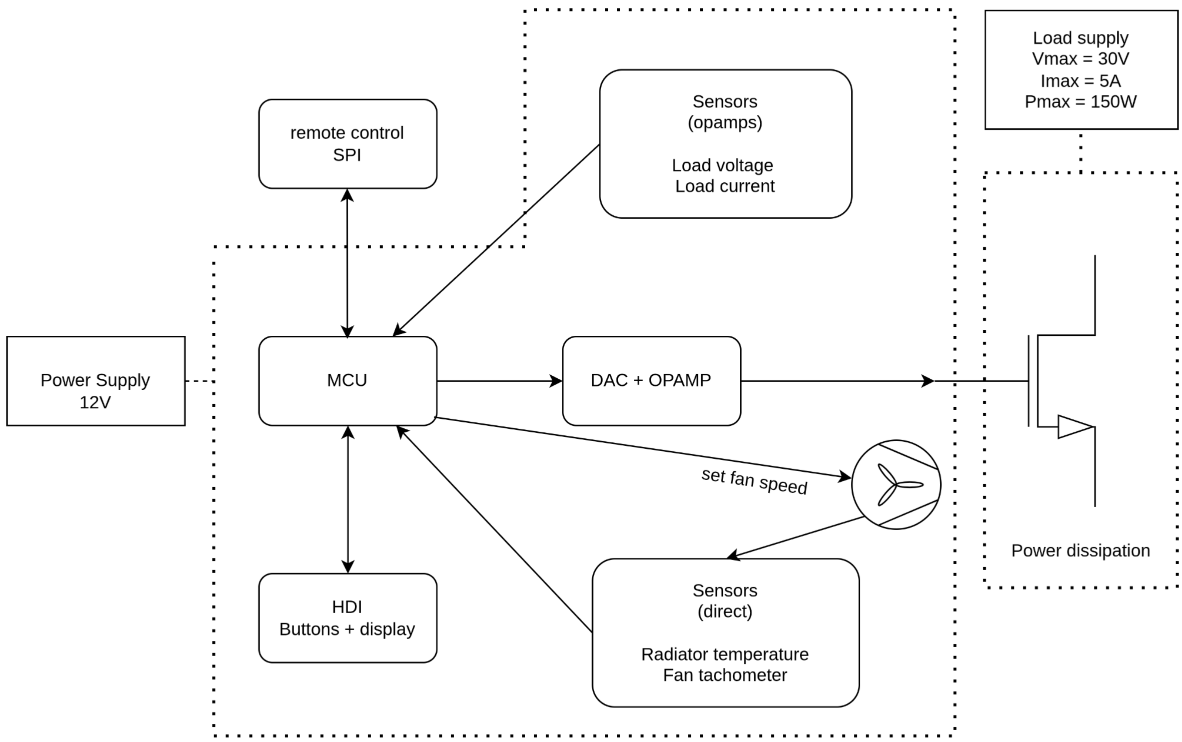

Figure 3 – System block diagram.

Main Features

- 150 W max power dissipation, 0-30V max, 0-5 A max
- Adjustable Constant Current (CC) mode
- Adjustable Constant Resistance (CR) mode
- Adjustable Constant Power (CP) mode
- Voltage and current measurement
- Temperature measurement and fan control
- Protection mechanisms (overcurrent, overtemperature, etc.)
- Simple human interface with a display, two push-buttons and a potentiometer.
- Communication interface using SPI

Human interface
1º - Press the Select Mode button to cycle through CC, CP and CR modes.
2º - Rotate the potentiometer to adjust the desired setpoint for the selected mode.
3º - Press the OK button to apply the new mode and setpoint.

Applications
• Power supply testing
• Battery discharge testing
• Solar panel characterization
• Educational laboratory experiments

SPI interface
Pinout: SCK, MOSI, MISO, CS, GND.

Data protocol
Command packet (12 bytes): sync (4), cmd (4), arg (4)
Status packet (20 bytes): sync (4), mode (4), setpoint (4), voltage (4), current (4)

More information is available in the firmware source code.
TODO: retransmit command packet on error.

Dimensions
Main board: 12 × 8 cm, Buttons board: 8 × 2 cm
Enclosure dimensions: 12 × 8 × 5 cm
Heatsink assembly dimensions: 15 × 6 × 7.5 cm.

Firmware
https://github.com/inaciose/progload

The following constants can be adjusted to calibrate the instrument and compensate for component tolerances.

- VOLTAGE_GAIN: Voltage measurement calibration factor.
- CURRENT_GAIN: Current measurement calibration factor.
- DAC_CAL: DAC output calibration factor.
- VCC_CAL: Internal ADC reference calibration constant.

Safety Notes
• The heatsink can reach high temperatures during operation.
• Do not exceed 30 V input voltage.
• Do not exceed 150 W power dissipation.
• Ensure adequate cooling.
• Never operate the device without the cooling fan.

Component list

Q	Description	Q	Description
1	Arduino nano 328P (MCU)	1	Capacitor 1 nF
1	MCP4725 (DAC)	1	Capacitor 22 nF
1	LM358	8	Capacitor 100 nF
1	MCP602 I/P	1	Capacitor 330 nF
1	LM7808	2	Capacitor 100 uF
3	IFRP260	2	1N4007
1	NTC 10k	3	Zener 16V
1	OLED	1	Resistor 0.1 (min 3W)
1	5x1 right-angle male header 90º	3	Resistor 0.1 (min 1W)
2	2-way terminal block	5	Resistor 150 Ω
2	3-way terminal block	1	Resistor 2k Ω
1	6x1 female header	1	Resistor 2.2 Ω
1	4x1 female header	4	Resistor 10k Ω
2	push buttons	2	Resistor 18k Ω
1	1k potentiometer	3	Resistor 20k Ω
1	Led	1	Resistor 100k Ω
1	Fan 5V w/pwm & tach	1	Resistor 180k Ω
		1	Resistor 330 Ω

Circuit diagram

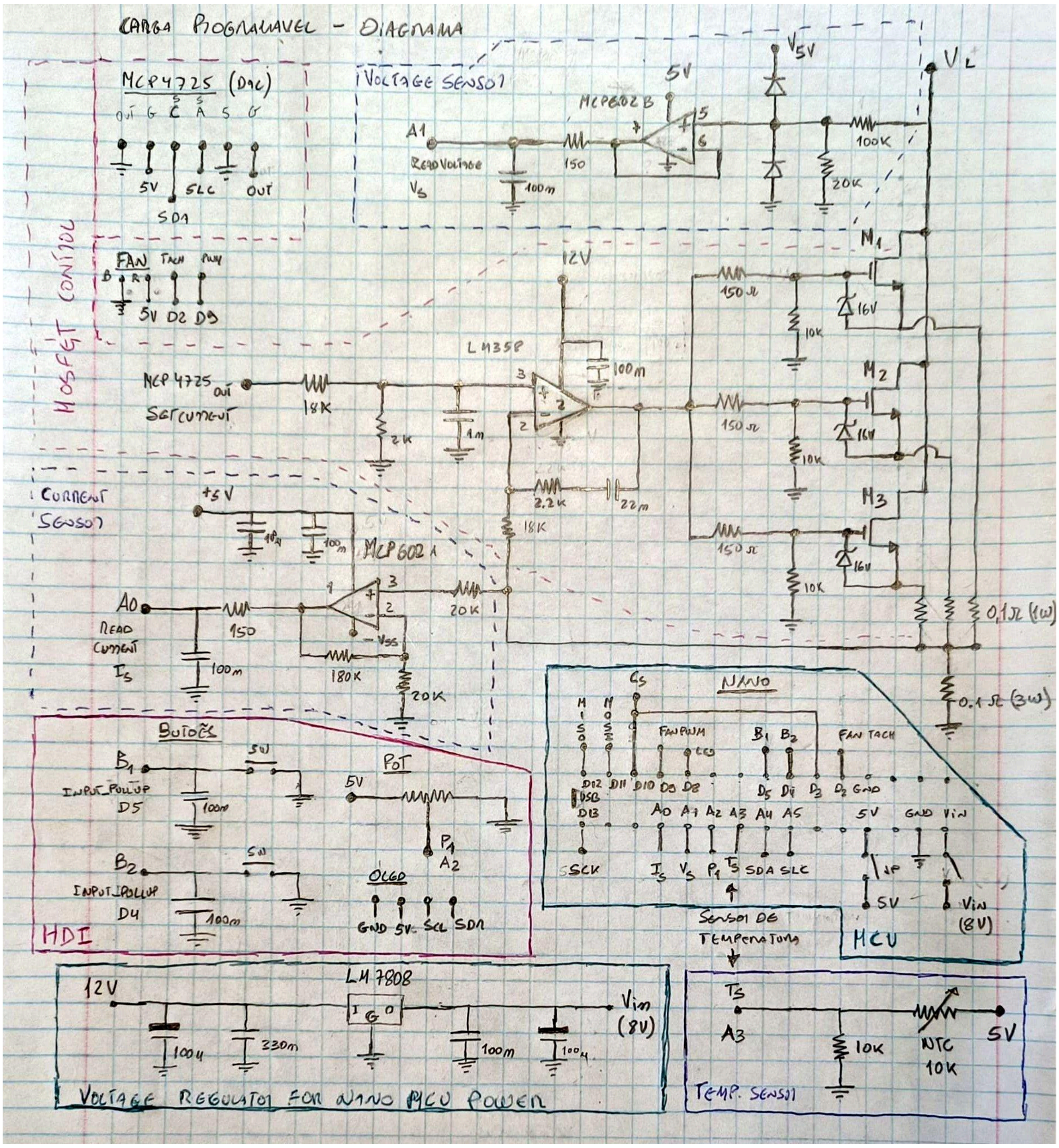

Figure 4 – Prototype circuit schematic.

PCB layout

Will be available in the project repository.